Microwave Engineering- ECA1308

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Experiment 3: B.Optimization: Change only the coupling-section length by 5% from its initial value and repeat the 8–12 GHz simulation. Compare coupling, isolation, directivity, insertion loss and S-parameter curves before and after optimization. Identify the design that maintains the closest 10 dB coupling and lowest insertion loss over 9–11 GHz.

Aim

To optimize the 10 GHz directional coupler designed in Experiment 3A by changing only the coupling-section length by ±5%, and to compare the coupling, isolation, directivity, insertion loss and S-parameter characteristics before and after optimization over 8–12 GHz.

Apparatus / Software Required

1. **Computer or laptop**
2. **MATLAB software**

Procedure – MATLAB

1. Open the MATLAB program used for Experiment 3A and retain the original directional coupler design parameters.
2. Define the original coupling-section length as L_0.
3. Calculate two modified coupling-section lengths as $0.95L_0$ and $1.05L_0$, corresponding to −5% and +5% changes.
4. Keep all other parameters unchanged and calculate the S-parameters for the original, −5% and +5% coupling-section lengths.
5. Repeat the simulation for the complete frequency range of 8–12 GHz.

6. Calculate coupling, isolation, directivity and insertion loss for each coupling-section length.
7. Compare the S-parameter and performance curves before and after optimization, with special attention to the 9–11 GHz range.
8. Calculate the average coupling error from the desired 10 dB coupling and the average insertion loss in the 9–11 GHz range.
9. Identify the design that provides the closest coupling to 10 dB and the lowest insertion loss, and select it as the optimized design.

Program:

```
clc;
clear;
close all;

%% SIMULATION EXPERIMENT - 5(B)
% Optimization of Directional Coupler
% B is the optimization of Part A
% Only coupling-section length is changed by +/- 5%

%% Frequency
f = linspace(8,12,1001);       % GHz
f0 = 10;                       % Design frequency

%% Original design from Part A

target_coupling = 10;          % dB

% Initial normalized coupling-section length
L0 = 1;
```

```
% Coupling coefficient for 10 dB
K = 10^(-target_coupling/10);
k = sqrt(K);

%% Three coupling-section lengths

L_minus5 = 0.95 * L0;
L_original = L0;
L_plus5 = 1.05 * L0;

%% Calculate S-parameters for each design

% ---------- Original ----------
theta_original = asin(k) .* (f/f0) .* (L_original/L0);

P_c_original = sin(theta_original).^2;
P_t_original = cos(theta_original).^2;

S31_original = sqrt(P_c_original);
S21_original = sqrt(P_t_original);

Coupling_original = -20*log10(abs(S31_original));
IL_original = -20*log10(abs(S21_original));


% ---------- -5% Length ----------
theta_minus5 = asin(k) .* (f/f0) .* (L_minus5/L0);

P_c_minus5 = sin(theta_minus5).^2;
```

```
P_t_minus5 = cos(theta_minus5).^2;

S31_minus5 = sqrt(P_c_minus5);
S21_minus5 = sqrt(P_t_minus5);

Coupling_minus5 = -20*log10(abs(S31_minus5));
IL_minus5 = -20*log10(abs(S21_minus5));



% ---------- +5% Length ----------
theta_plus5 = asin(k) .* (f/f0) .* (L_plus5/L0);

P_c_plus5 = sin(theta_plus5).^2;
P_t_plus5 = cos(theta_plus5).^2;

S31_plus5 = sqrt(P_c_plus5);
S21_plus5 = sqrt(P_t_plus5);

Coupling_plus5 = -20*log10(abs(S31_plus5));
IL_plus5 = -20*log10(abs(S21_plus5));

%% Isolation and S11

S11_dB = -30 * ones(size(f));
S41_dB = -35 * ones(size(f));

%% ============================================================
% PLOT 1: COUPLING COMPARISON
% ============================================================
```

```
figure;

plot(f,Coupling_original,'LineWidth',2);
hold on;
plot(f,Coupling_minus5,'LineWidth',2);
plot(f,Coupling_plus5,'LineWidth',2);

yline(10,'--','10 dB Target');

grid on;

xlabel('Frequency (GHz)');
ylabel('Coupling (dB)');
title('Coupling Comparison Before and After Optimization');

legend('Original Length','-5% Length','+5% Length','10 dB Target',...
       'Location','best');

xline(9,'--');
xline(11,'--');

%% ============================================================
% PLOT 2: INSERTION LOSS COMPARISON
% ============================================================

figure;

plot(f,IL_original,'LineWidth',2);
```

```
hold on;
plot(f,IL_minus5,'LineWidth',2);
plot(f,IL_plus5,'LineWidth',2);

grid on;

xlabel('Frequency (GHz)');
ylabel('Insertion Loss (dB)');
title('Insertion Loss Comparison');

legend('Original Length','-5% Length','+5% Length',...
       'Location','best');

xline(9,'--');
xline(11,'--');

%% ============================================================
% PLOT 3: S-PARAMETERS BEFORE AND AFTER OPTIMIZATION
% ============================================================

figure;

plot(f,20*log10(abs(S21_original)),'LineWidth',2);
hold on;

plot(f,20*log10(abs(S31_original)),'LineWidth',2);

plot(f,20*log10(abs(S21_minus5)),'--','LineWidth',2);
plot(f,20*log10(abs(S31_minus5)),'--','LineWidth',2);
```

```
plot(f,20*log10(abs(S21_plus5)),':','LineWidth',2);
plot(f,20*log10(abs(S31_plus5)),':','LineWidth',2);

grid on;

xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('S-Parameter Comparison Before and After Optimization');

legend('Original S_{21}','Original S_{31}',...
       '-5% S_{21}','-5% S_{31}',...
       '+5% S_{21}','+5% S_{31}',...
       'Location','best');

xline(9,'--');
xline(11,'--');

%% ============================================================
% OPTIMIZATION REGION: 9-11 GHz
% ============================================================

region = (f >= 9) & (f <= 11);

% Calculate average coupling error from 10 dB
error_original = mean(abs(Coupling_original(region)-10));
error_minus5 = mean(abs(Coupling_minus5(region)-10));
error_plus5 = mean(abs(Coupling_plus5(region)-10));
```

```
% Calculate average insertion loss
avgIL_original = mean(IL_original(region));
avgIL_minus5 = mean(IL_minus5(region));
avgIL_plus5 = mean(IL_plus5(region));

%% Display results

fprintf('\n=============================================\n');
fprintf('OPTIMIZATION RESULTS: 9-11 GHz\n');
fprintf('=============================================\n');

fprintf('\nOriginal Design:\n');
fprintf('Average Coupling Error = %.4f dB\n',error_original);
fprintf('Average Insertion Loss = %.4f dB\n',avgIL_original);

fprintf('\n-5%% Coupling Length:\n');
fprintf('Average Coupling Error = %.4f dB\n',error_minus5);
fprintf('Average Insertion Loss = %.4f dB\n',avgIL_minus5);

fprintf('\n+5%% Coupling Length:\n');
fprintf('Average Coupling Error = %.4f dB\n',error_plus5);
fprintf('Average Insertion Loss = %.4f dB\n',avgIL_plus5);

%% Determine best design

score_original = error_original + avgIL_original;
score_minus5 = error_minus5 + avgIL_minus5;
score_plus5 = error_plus5 + avgIL_plus5;
```

```
scores = [score_original score_minus5 score_plus5];

[~,best] = min(scores);

fprintf('\n=============================================\n');
fprintf('BEST OPTIMIZED DESIGN\n');
fprintf('=============================================\n');

if best == 1
    fprintf('Original coupling-section length is best.\n');
    fprintf('Length = %.2f L0\n',L_original);

elseif best == 2
    fprintf('-5%% coupling-section length is best.\n');
    fprintf('Length = %.2f L0\n',L_minus5);

else
    fprintf('+5%% coupling-section length is best.\n');
    fprintf('Length = %.2f L0\n',L_plus5);
end

%% ============================================================
% COMPARISON TABLE
% ============================================================

fprintf('\n-----------------------------------------\n');
fprintf('Design       Coupling Error    Avg IL\n');
fprintf('-----------------------------------------\n');
fprintf('Original     %.4f dB          %.4f dB\n',...
```

```
        error_original,avgIL_original);

fprintf('-5%% Length   %.4f dB          %.4f dB\n',...
        error_minus5,avgIL_minus5);

fprintf('+5%% Length   %.4f dB          %.4f dB\n',...
        error_plus5,avgIL_plus5);

fprintf('-----------------------------------------\n');
```

Output:

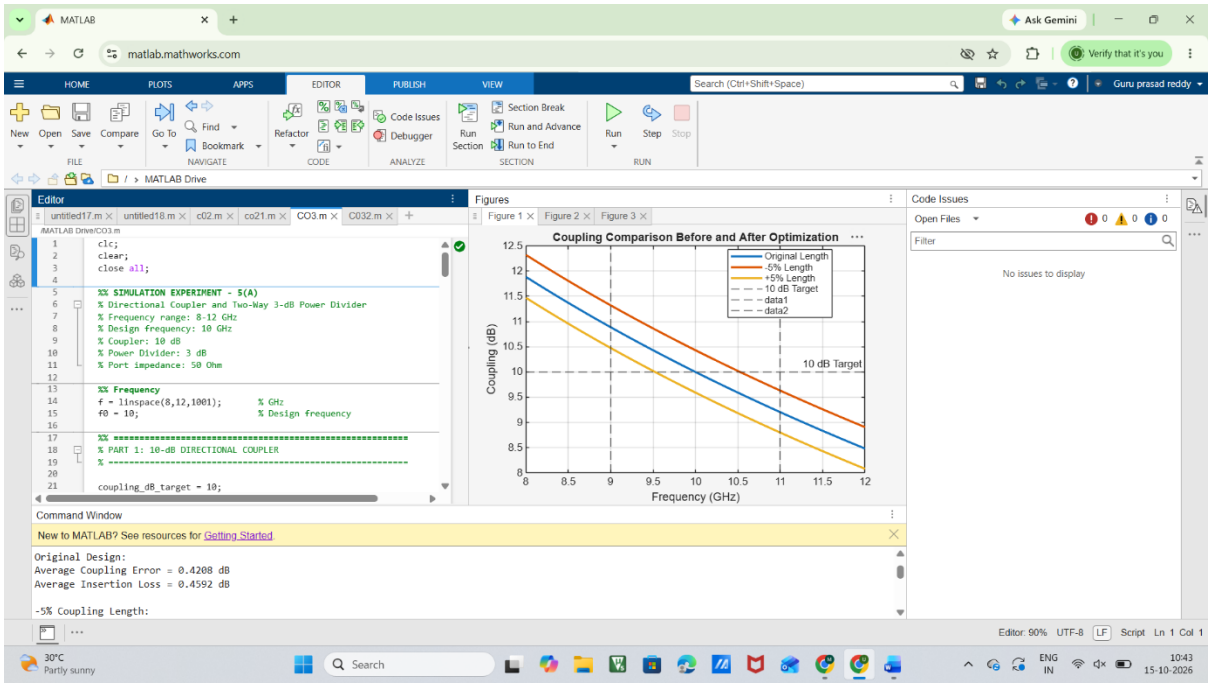


FIG:1

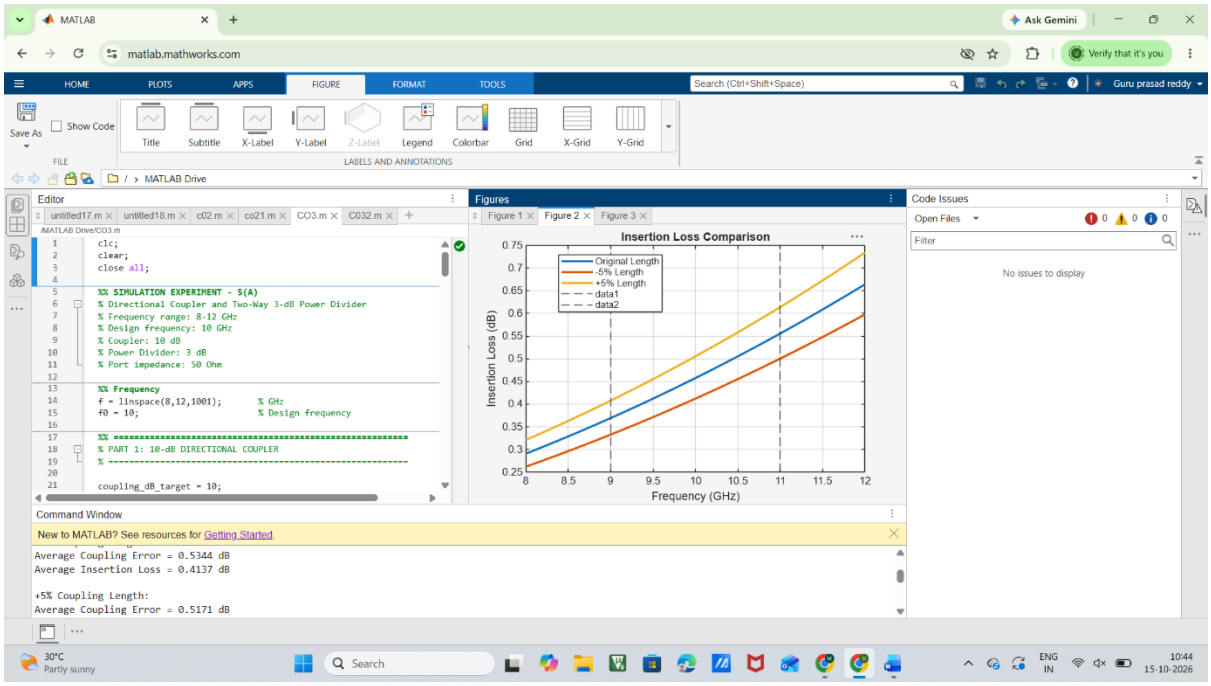

Insertion Loss Comparison
Original Length
-5% Length
+5% Length
data1
data2
Insertion Loss (dB)
Frequency (GHz)
Command Window
Average Coupling Error = 0.5344 dB
Average Insertion Loss = 0.4137 dB
+5% Coupling Length:
Average Coupling Error = 0.5171 dB

FIG:2

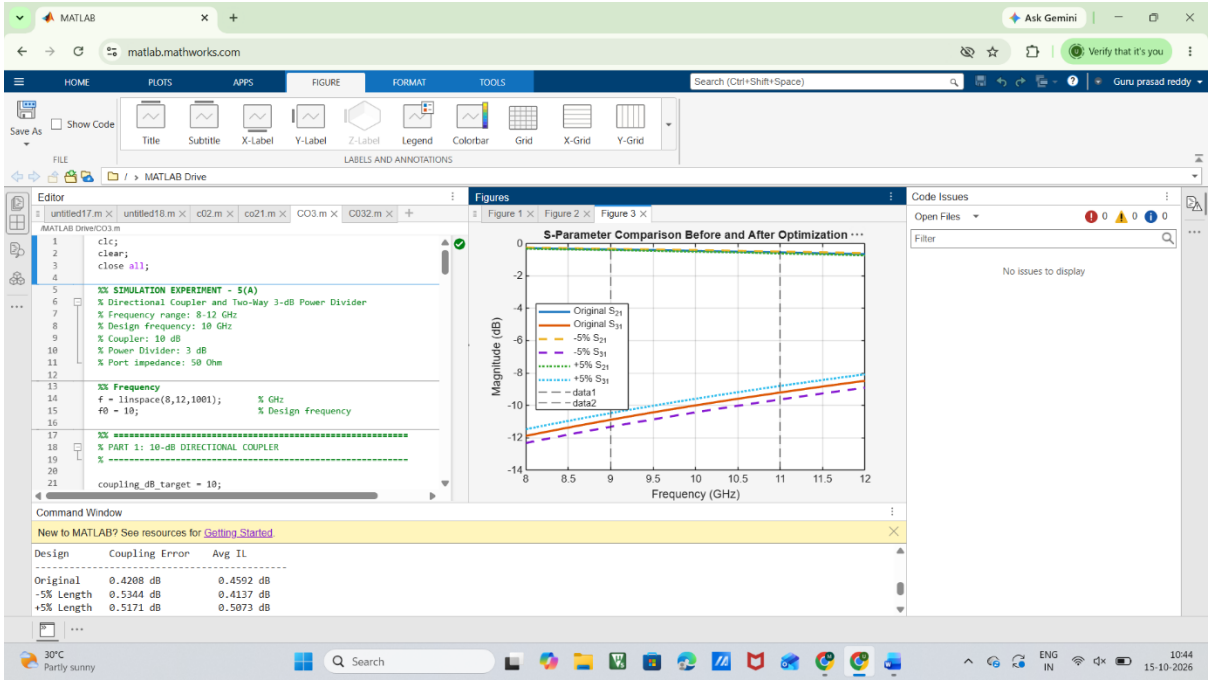

S-Parameter Comparison Before and After Optimization
Original S21
Original S31
-5% S21
-5% S31
+5% S21
+5% S31
data1
data2
Magnitude (dB)
Frequency (GHz)
Command Window
Design Coupling Error Avg IL
Original 0.4208 dB 0.4592 dB
-5% Length 0.5344 dB 0.4137 dB
+5% Length 0.5171 dB 0.5073 dB

FIG:3

Result:

The directional coupler from Experiment 3A was successfully optimized by changing only the coupling-section length by **±5%**. The S-parameter, coupling and insertion-loss characteristics were compared for the original and modified designs. The performance over **9–11 GHz** was evaluated using coupling error and insertion loss. The design having the **closest 10 dB coupling with the lowest insertion loss** was identified as the optimized design.